5. Limit Theorems

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SS26 MTH344: Mathematical Statistics

ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

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Lecture Topics

- This lecture reviews the asymptotic behavior of the sample mean,

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{1}$$

  where $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables, with a particular focus on the Law of Large Numbers and the Central Limit Theorem.
- See Chapter 5 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.

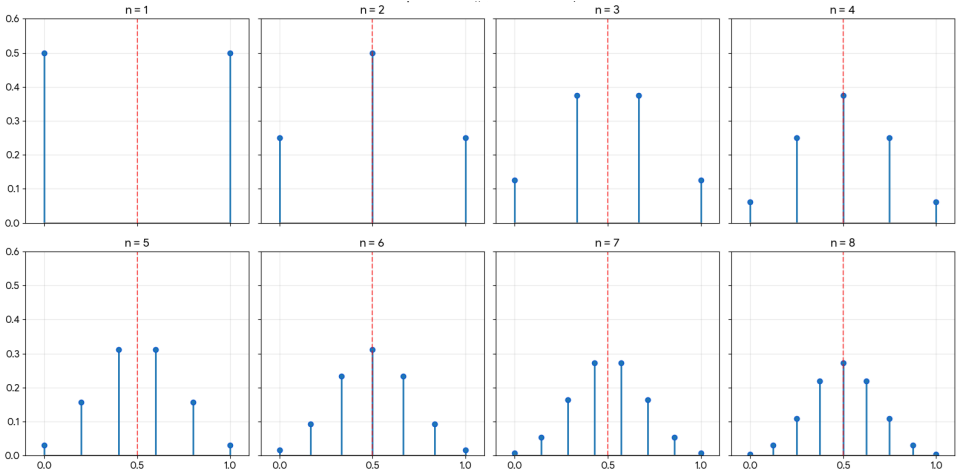
Introduction

- As a motivating example, consider tossing a fair coin n times. Let $X_i \sim \mathrm{Ber}(1/2)$ denote the number of heads in the i-th trial. Then, the sample mean of the heads can be expressed as $\bar{X}_n$.
- How can we relate the sample mean $\bar{X}_n$ to the population mean $\mu = 1/2$?
- One of the well-known relations between $\bar{X}_n$ and μ is given by:

$$\mathrm{E}[\bar{X}_n] = \mu. \tag{2}$$

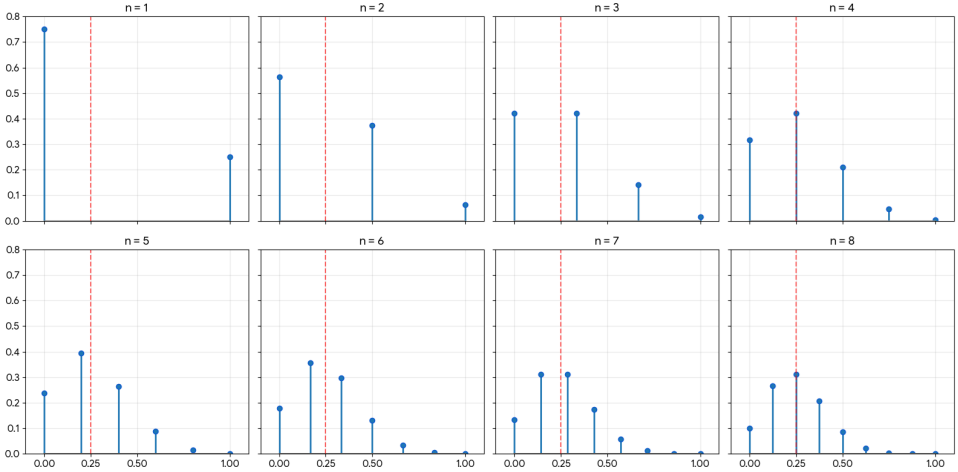
This implies that a weighted average of the possible values of $\bar{X}_n$ is μ, but this relation still allows the values of $\bar{X}_n$ to be substantially spread out around μ.

Introduction



PMF of $\bar{X}_n$ where $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(0.5)$.

Introduction



PMF of $\bar{X}_n$ where $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(0.25)$.

Introduction

- While $\bar{X}_n$ exhibits variability, its spread decreases as n increases.
- The Law of Large Numbers states that this variability vanishes asymptotically.
- The Central Limit Theorem establishes that the standardized $\bar{X}_n$,

$$\frac{\bar{X}_n - \mathrm{E}[\bar{X}_n]}{\sqrt{\mathrm{Var}[\bar{X}_n]}}, \tag{3}$$

  converges in distribution to the standard Gaussian distribution.
- Both results hold for arbitrary distributions (given some finite moments) and are fundamental to Statistics and Probability.

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The Law of Large Numbers

- How should we define the convergence of a random variable $\bar{X}_n$ to a point μ?
- As a naive idea, how about defining the convergence as:

$$P(\bar{X}_n = \mu) \to 1 \quad \text{as } n \to \infty \tag{4}$$

- **Q**: Does this hold in the example of tossing a fair coin? What about for continuous random variables?

The Law of Large Numbers

- Note that $\bar{X}_n : S \to \mathbb{R}$ is a function. Thus, $\left(\bar{X}_n(\cdot)\right)_{n\geq 1}$ can be viewed from two perspectives:
  1. as a sequence of values for a fixed s (pointwise-level view)
  2. as a sequence of functions (function-level view)

  In this context, we can define its convergence at the pointwise level and the functional level.
  1. (Strong convergence)

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = P(\{s : \lim_{n\to\infty} \bar{X}_n(s) = \mu\}) = 1. \tag{5}$$

  2. (Weak convergence) For any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \to 0 \quad \text{as } n \to \infty. \tag{6}$$

- In this context, the Law of Large Numbers has two versions: (i) the Strong Law of Large Numbers and (ii) the Weak Law of Large Numbers.

The Law of Large Numbers

- **Strong Law of Large Numbers**: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with $\mathrm{E}[X_i] = \mu$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Then,

$$P(\{s : \lim_{n \to \infty} \bar{X}_n(s) = \mu\}) = 1. \tag{7}$$

- That is, with probability 1, the sequence $\bar{X}_n$ converges to μ. This convergence is denoted by:

$$\bar{X}_n \overset{a.s.}{\to} \mu, \tag{8}$$

  which is read as '$\bar{X}_n$ converges to μ almost surely (a.s.)'.
- The proof of this result is beyond the scope of this course. A simplified proof assuming a finite fourth moment can be found in Section 8.4 of *A First Course in Probability* [FCP] (Ross, 2020).

The Law of Large Numbers

- **Theorem 5.2.A (Weak Law of Large Numbers)**: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with $\mathrm{E}[X_i] = \mu$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Then, for any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad \text{as } n \to \infty. \tag{9}$$

- That is, as n increases, the probability of observing $\bar{X}_n$ deviating substantially from μ becomes negligible. This convergence is denoted by:

$$\bar{X}_n \xrightarrow{p} \mu, \tag{10}$$

which is read as '$\bar{X}_n$ converges to μ in probability'.

The Law of Large Numbers

Exercise 1

Prove Markov's inequality: If X is a random variable that takes only non-negative values, then for any value $a > 0$,

$$P(X \geq a) \leq \frac{\mathrm{E}[X]}{a}. \tag{11}$$

The Law of Large Numbers

Exercise 2

Prove Chebyshev's inequality: If X is a random variable with finite mean μ and variance σ^2, then for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \sigma^2/k^2. \tag{12}$$

The Law of Large Numbers

Exercise 3

Prove the Weak Law of Large Numbers assuming finite variance: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Then, for any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad \text{as } n \to \infty. \tag{13}$$

Application of the Weak Law of Large Numbers

- **Example 5.2.A (Monte Carlo Integration)**: Let $U_1, U_2, \ldots$ be a sequence of i.i.d. random variables such that $U_i \sim \mathcal{U}(0,1)$. Then, for any integrable function f (i.e., $\int_0^1 |f(u)|du < \infty$),

$$\frac{1}{n}\sum_{i=1}^{n} f(U_i) \xrightarrow{p} \int_0^1 f(u)du. \tag{14}$$

- Monte Carlo integration is particularly useful when the integral has no closed-form, e.g.,

$$\Phi(1) - \Phi(0) = \int_0^1 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \approx \frac{1}{n}\sum_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{U_i^2}{2}\right). \tag{15}$$

Since the integral cannot be solved analytically, we approximate it using the sample mean above.

Application of the Weak Law of Large Numbers

- **Example 5.2.B (Repeated Measurements)**: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mathrm{E}[X_i] = \mu$. Similar to the Monte Carlo example, these can be viewed as repeated measurements of a quantity μ.
- With a finite sample size n, how do we evaluate the estimation error of $\bar{X}_n$? If $\mathrm{Var}[X_i] = \sigma^2$ is finite, a natural measure of this error is:

$$\mathrm{Var}[\bar{X}_n] = \mathrm{E}[(\bar{X}_n - \mu)^2] = \frac{\sigma^2}{n}. \tag{16}$$

- When σ^2 is unknown, we can estimate it as follows:

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2 \xrightarrow{p} \mathrm{E}[X_i^2] - \big(\mathrm{E}[X_i]\big)^2 = \sigma^2. \tag{17}$$

See Practice Problem 3 and the next Exercise for details.

Application of the Weak Law of Large Numbers

Exercise 4

Let $X_n \xrightarrow{p} \alpha$ and $Y_n \xrightarrow{p} \beta$ where $(\alpha, \beta)^\top \in \mathbb{R}^2$.

a) Show that $X_n + Y_n \xrightarrow{p} \alpha + \beta$

b) Show that $X_n Y_n \xrightarrow{p} \alpha\beta$

Simulation: Weak Law of Large Numbers

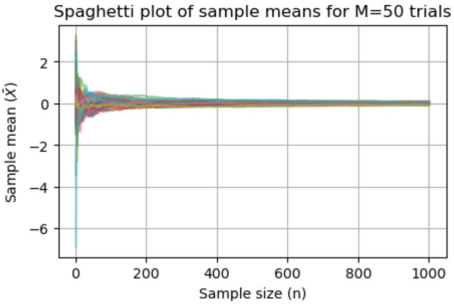
- **Experiment:** We examine whether $P(|\bar{X}_n - \mu| > \epsilon)$ converges to zero.
  1. Generate $M = 50$ sample paths of size $n = 1000$. Draw a "spaghetti plot" consisting of these M trajectories of $\bar{X}_n$ (x-axis: n; y-axis: $\bar{X}_n$).
  2. Perform $M = 1000$ independent simulations of sample size $n = 1000$. Plot the empirical estimates (relative frequencies) of $P(|\bar{X}_n - \mu| > 0.1)$ as a function of n.
- **Distribution:** Let X follow a Student's t-distribution[1] with ν degrees of freedom:

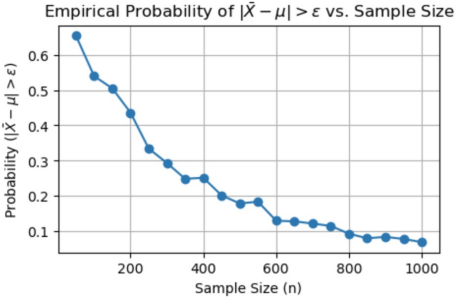
$$f(x;\nu) \propto \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$$

  - The mean is $\mu = 0$ if $\nu > 1$ (undefined otherwise).
  - The variance is finite if and only if $\nu > 2$.
- A Python implementation of this simulation can be found here (GitHub Link).

[1] We will introduce this distribution in detail in Chapter 6.

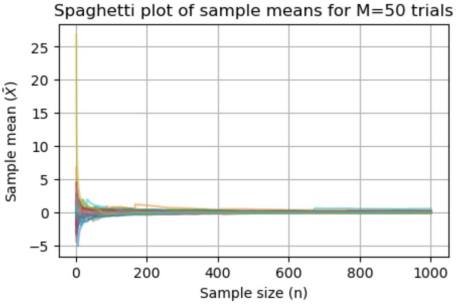
Simulation: Weak Law of Large Numbers

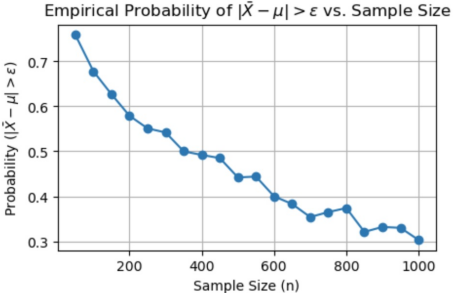




- Results when $\nu = 3$. In this case, $\mu = \mathrm{E}[X_i] = 0$ and $\mathrm{Var}[X_i] = 3$.

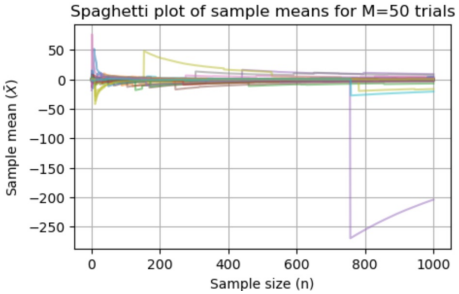
Simulation: Weak Law of Large Numbers

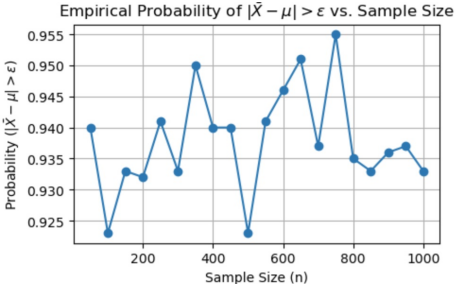




- Results when $\nu = 2$. In this case, $\mu = \mathrm{E}[X_i] = 0$ and $\mathrm{Var}[X_i] = \infty$.
- Given $n = 1000$, the empirical estimate of $P(|\bar{X}_n - \mu| > \epsilon)$ increased a lot. We need larger sample sizes to empirically confirm the convergence of $\bar{X}_n$.

Simulation: Weak Law of Large Numbers





- Results when $\nu = 1$. In this case, $\mu = \mathrm{E}[X_i]$ is undefined. This distribution is referred to as the *Cauchy distribution*. In the right figure, $\mu = 0$ was chosen for consistent visualization.
  Q: Based on plots, should we conclude that $\bar{X}_n$ does not converge to zero (or any other point), or could this result be due to a small sample size issue?

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Statistical Inference with Sample Mean

- While the Weak Law of Large Numbers guarantees that

$$\bar{X}_n \xrightarrow{p} \mu, \tag{18}$$

  statistical inference often requires understanding the full distribution of $\bar{X}_n$.
- Suppose we want to test if a coin is fair. After flipping the coin 100 times, we observe 40 heads (relative frequency 0.4).
- Can we conclude that the true probability of heads is less than 0.5? Although $0.4 < 0.5$, this deviation could simply be due to randomness.
- A standard approach in statistics is to calculate the probability of observing a result at least as extreme as the actual observation, assuming the coin is fair (this is the *p-value*):

$$P(\bar{X}_n \leq 0.4). \tag{19}$$

Statistical Inference with Sample Mean

- Let $X_1, \ldots, X_n$ be i.i.d. random variables from Ber(0.5). Note that the relative frequency is the sample mean $\bar{X}_n$. For $n = 100$:

$$P(\bar{X}_{100} \leq 0.4) = \sum_{i=0}^{40} \binom{100}{i} 0.5^i (1 - 0.5)^{100-i} = 1.76\%. \tag{20}$$

- Thus, in a hypothetical scenario where we infinitely repeat the experiment of tossing a fair coin 100 times, we would observe $\bar{X}_{100} \leq 0.4$ in 1.76% of the cases.
- What if we do not have a closed-form expression for the CDF of $\bar{X}_n$?

Statistical Inference with Sample Mean

- Remarkably, the **Central Limit Theorem (CLT)** says that, as long as $\sigma^2 < \infty$,

$$P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z\right) \to \Phi(z) \quad \text{as } n \to \infty, \tag{21}$$

  where $\Phi(\cdot)$ is the CDF of the standard normal distribution, $\mathcal{N}(0,1)$.
- For example, with the CLT, we can approximate $P(\bar{X}_{100} \leq 0.4)$ without a closed-form expression for the CDF of $\bar{X}_n$:

$$\begin{aligned} P\left(\bar{X}_{100} \leq 0.4\right) &= P\left(\frac{\sqrt{100}\left(\bar{X}_{100} - 0.5\right)}{\sqrt{0.5(1-0.5)}} \leq \frac{\sqrt{100}\,(0.4-0.5)}{\sqrt{0.5(1-0.5)}}\right) \\ &\approx \Phi\left(\frac{\sqrt{100}\,(0.4-0.5)}{\sqrt{0.5(1-0.5)}}\right) \\ &= \Phi\left(-2\right) = 2.28\% \end{aligned} \tag{22}$$

Convergence in Distribution

- Before studying the CLT, we first introduce the third type of convergence of random variables: *convergence in distribution*.
- **Convergence in Distribution**: Let $X_1, X_2, \ldots$ be a sequence of random variables with CDFs $F_1, F_2, \ldots$, and let X be a random variable with CDF F. We say that X_n converges in distribution to X if:

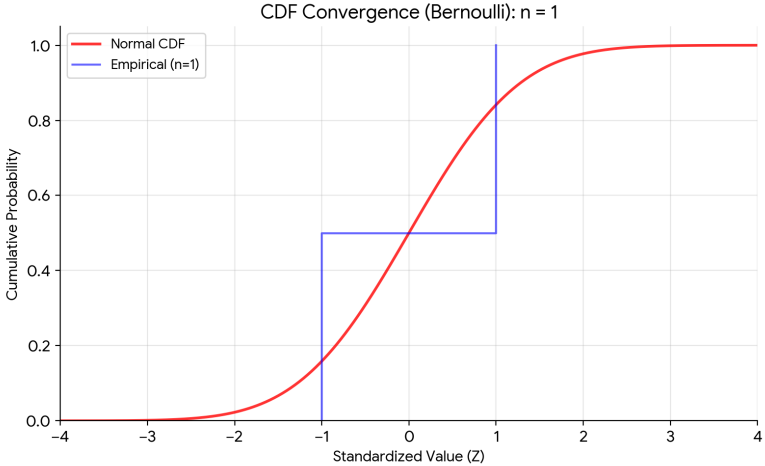
$$\lim_{n\to\infty} F_n(x) = F(x) \tag{23}$$

at every point x at which F is continuous. This convergence is denoted by:

$$X_n \xrightarrow{d} X, \tag{24}$$

which is read as 'X_n converges to X in distribution'.
- For a sequence of random vectors $\boldsymbol{X}_n$ and a random vector $\boldsymbol{X}$, we say that $\boldsymbol{X}_n \xrightarrow{d} \boldsymbol{X}$ if $\lim_{n\to\infty} F_{\boldsymbol{X}_n}(\boldsymbol{x}_n = \boldsymbol{x}) = F_{\boldsymbol{X}}(\boldsymbol{x})$ for every $\boldsymbol{x}$ at which F is continuous.

Convergence in Distribution



Convergence in Distribution

Exercise 5

Slutsky's Theorem: Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, where c is a constant and X has a continuous CDF.[a]

a) Show that $X_n + Y_n \xrightarrow{d} X + c$

b) Show that $X_n Y_n \xrightarrow{d} cX$

[a] The theorem holds generally without assuming the continuity of the CDF of X.

Convergence in Distribution

The Central Limit Theorem

- The MGF serves as an essential tool in deriving the CLT.
- The convergence of MGFs implies the convergence of the corresponding CDFs. This result is formulated as the *Continuity Theorem*, which we state here without proof.
- **Theorem 5.3.A (Continuity Theorem)**: Let $(F_n)_{n\geq 1}$ be a sequence of CDFs with corresponding MGFs $(M_n)_{n\geq 1}$. Let F be a CDF with the MGF M. If $M_n(t) \to M(t)$ for all t in an open interval containing zero, then $F_n(x) \to F(x)$ at all continuity points of F.

The Central Limit Theorem

- **Theorem 5.3.B (Central Limit Theorem)**: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with mean μ, variance σ^2. Then,

$$\lim_{n\to\infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \Phi(x), \quad -\infty < x < \infty. \tag{25}$$

  That is, $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z \sim \mathcal{N}(0,1)$.
- The CLT holds generally without assuming the existence of the MGF, but we assume this condition here to simplify the proof.

The Central Limit Theorem

Exercise 6

Prove the CLT assuming the existence of MGFs: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with mean μ, variance σ^2, common CDF F, and MGF M defined in a neighborhood of zero. Then,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z \sim \mathcal{N}(0,1). \tag{26}$$

The Central Limit Theorem

The Central Limit Theorem

- **Multivariate Central Limit Theorem**: Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ be a sequence of i.i.d. random vectors in $\mathbb{R}^d$ with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ. Then,

$$\sqrt{n}(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \boldsymbol{Z}, \quad \boldsymbol{Z} \sim \mathcal{N}(0, \Sigma). \tag{27}$$

- For example, denoting the k-th moment by $m_k = \mathrm{E}[X^k]$ and assuming $m_4 < \infty$,

$$\sqrt{n}\left(\begin{pmatrix} \bar{X} \\ \overline{X^2} \end{pmatrix} - \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}\right) \xrightarrow{d} \boldsymbol{Z} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} m_2 - m_1^2 & m_3 - m_1 m_2 \\ m_3 - m_1 m_2 & m_4 - m_2^2 \end{pmatrix}\right). \tag{28}$$

Application of The Central Limit Theorem

- **Example 5.3.C**: Let $U_i \overset{\text{i.i.d.}}{\sim} \mathcal{U}[0,1]$ where $\mathrm{E}[U_i] = 1/2$ and $\mathrm{Var}[U_i] = 1/12$. Then, by the CLT,

$$\frac{\bar{U}_n - 1/2}{\sqrt{1/12}/\sqrt{n}} = \sqrt{12n}\left(\bar{U}_n - 1/2\right) \to Z \sim \mathcal{N}(0, 1^2). \tag{29}$$

- When $n = 12$, this standardized sample mean is equal to:

$$\sum_{i=1}^{12} U_i - 6, \tag{30}$$

which was commonly used for generating Gaussian random variables in earlier algorithms.

Application of The Central Limit Theorem

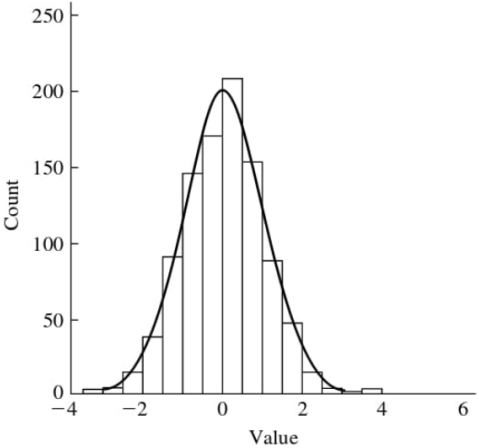


Fig. 5.1. [MSDA]: A histogram of 1000 samples of $\sum_{i=1}^{12} U_i - 6$.

Application of The Central Limit Theorem

- **Example 5.3.E (Measurement Error)**: Let $X_1, \ldots, X_n$ be i.i.d. random variables with $\mathrm{E}[X_i] = \mu$. They can be viewed as repeated measurements of a quantity μ.
- We can apply Chebyshev's inequality and the CLT to derive the required sample size n that satisfies:

$$P(|\bar{X}_n - \mu| < c) \geq \gamma \tag{31}$$

where $c > 0$ represents the admissible tolerance level of error and $0 < \gamma < 1$ represents the reliability level.
- Using Chebyshev's Inequality, we have:

$$P(|\bar{X}_n - \mu| < c) \geq 1 - \frac{\sigma^2}{nc^2}. \tag{32}$$

Here, the RHS is greater than or equal to γ if and only if:

$$n \geq \frac{\sigma^2}{c^2} \frac{1}{1-\gamma}. \tag{33}$$

Application of The Central Limit Theorem

- In contrast, using the CLT, we have:

$$P(|\bar{X}_n - \mu| < c) \approx \Phi\left(\frac{c\sqrt{n}}{\sigma}\right) - \Phi\left(-\frac{c\sqrt{n}}{\sigma}\right) \tag{34}$$

Here, the RHS is greater than or equal to γ if and only if:

$$n \geq \frac{\sigma^2}{c^2}\left(\Phi^{-1}\left(\frac{1+\gamma}{2}\right)\right)^2. \tag{35}$$

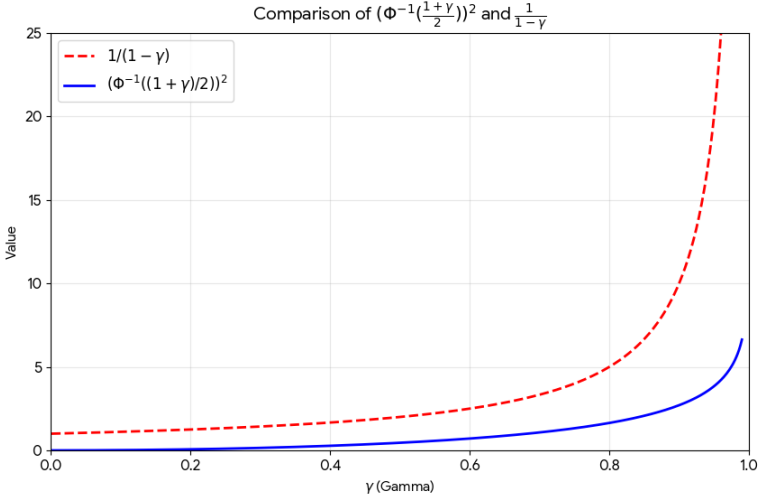
Application of The Central Limit Theorem

Exercise 7

Show that, for a given $c > 0$ and $0 < \gamma < 1$, $\Phi\left(\frac{c\sqrt{n}}{\sigma}\right) - \Phi\left(-\frac{c\sqrt{n}}{\sigma}\right) \geq \gamma$ holds if and only if

$$n \geq \frac{\sigma^2}{c^2}\left(\Phi^{-1}\left(\frac{1+\gamma}{2}\right)\right)^2 \tag{36}$$

Application of The Central Limit Theorem



Q: What would be the pros and cons of the two approaches?

Simulation: Central Limit Theorem

- **Experiment:** We will examine whether $P(\sqrt{n}(\bar{X}_n - \mu)/\sigma \leq z)$ converges to $\Phi(z)$:
  1. Generate $M = 1000$ independent datasets, each of size $n \in \{25, 50, 100, 200\}$, to compute M realizations of the standardized sample mean $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ for each n.
  2. Examine whether the empirical distribution functions of these realizations converge to the standard normal CDF, $\Phi(z)$.
- **Distribution:** We consider a Gaussian Mixture Model (GMM) with PDF:

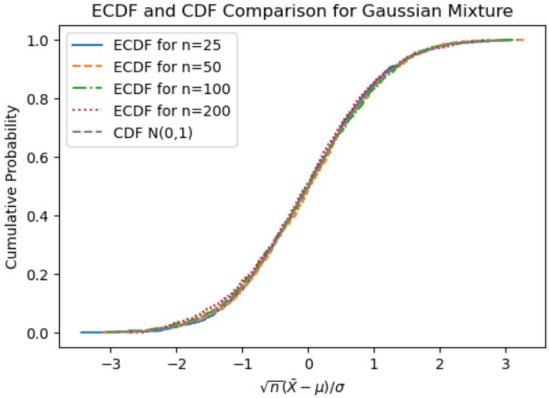
$$f_\theta(x) = \pi\mathcal{N}(x; \mu_1, \sigma_1^2) + (1 - \pi)\mathcal{N}(x; \mu_2, \sigma_2^2), \tag{37}$$

where $\theta := (\pi, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2)^\top$ and ϕ denotes the normal PDF. For θ, we use:

$$\theta = (0.5, 0, 1^2, 3, 1.5^2)^T \tag{38}$$

- A Python implementation of this simulation can be found here (GitHub Link).

Simulation: Central Limit Theorem

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MISE for n=25: 2.926529079375994e-05
MISE for n=50: 4.158184313277996e-05
MISE for n=100: 2.709211076327643e-05
MISE for n=200: 0.00012699633237750485
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Practice Problem I

1. Problems 5, 21, 22, and 29 in Chapter 5 in [MSDA].
2. Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with $\mathrm{E}[X_i] = \mu$. For a given $\epsilon > 0$, let $A_{k,\epsilon} = \{\omega : |\bar{X}_k - \mu| \leq \epsilon\}$. Show that, for any $\omega \in \Omega$ such that $\lim_{n\to\infty} \bar{X}_n(\omega) = \mu$, $\omega \in \bigcap_{k=n}^{\infty} A_{k,\epsilon}$ for a sufficiently large $n(w) \in \mathbb{N}$.
3. Let $X_n \xrightarrow{p} \alpha$ where $\alpha \in \mathbb{R}$, and let g be a function continuous at α. Show that

$$g(X_n) \xrightarrow{p} g(\alpha). \tag{39}$$

4. Let $X_n \xrightarrow{a.s.} \alpha$ and $Y_n \xrightarrow{a.s.} \beta$ where $(\alpha, \beta)^\top \in \mathbb{R}^2$.
   a) Prove or provide a counterexample: $X_n + Y_n \xrightarrow{a.s.} \alpha + \beta$
   b) Prove or provide a counterexample: $X_n Y_n \xrightarrow{a.s.} \alpha\beta$
5. For a given constant c, $X_n \xrightarrow{p} c$ if and only if $X_n \xrightarrow{d} c$.
6. Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$.

Practice Problem II

a) Prove or provide a counterexample: $X_n + Y_n \xrightarrow{d} X + Y$

b) Prove or provide a counterexample: $X_n Y_n \xrightarrow{d} XY$

7. Prove or provide a counterexample: Suppose that $(X_n, Y_n)^T \xrightarrow{d} (X, Y)^T$, and $X_n \perp\!\!\!\perp Y_n$ for all n. Then, $X \perp\!\!\!\perp Y$.
8. Prove or provide a counterexample: Suppose that $(X_n, Y_n)^T \xrightarrow{d} (X, Y)^T$ and X_n and Y_n are dependent for all n. Then, X and Y are also dependent.
9. Let $\phi(x)$ and $\Phi(x)$ be the PDF and CDF of the standard Gaussian distribution, respectively.

   a) *Mill's ratio* of the standard Gaussian is defined as $m(x) = (1 - \Phi(x))/\phi(x)$. For any $x > 0$, show that:

$$1 - \frac{1}{x^2} < xm(x) < 1. \tag{40}$$

Practice Problem III

b) Let $g(x) = x\phi(x)$. For any $x > 0$, show that:

$$g(x) \leq \frac{1}{\sqrt{2\pi}} e^{-1/2}. \tag{41}$$

Here, the RHS is approximately 0.242.

c) For any $\sigma^2 > 0$, $c > 0$, and $0 < \gamma < 1$, show that:

$$\frac{\sigma^2}{c^2}\left(\Phi^{-1}\left(\frac{1+\gamma}{2}\right)\right)^2 < \frac{\sigma^2}{c^2}\frac{1}{1-\gamma}. \tag{42}$$

That is, the CLT-based approach in Equation (34) is sharper than the approach based on Chebyshev's inequality in Equation (32).

Advanced Practice Problem I

1. Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables from $\mathrm{Exp}(\lambda)$, and let $X_{(i)}$ denote the i-th order statistic.
   a) Show that $Z_1 = nX_{(1)}$, $Z_2 = (n-1)(X_{(2)} - X_{(1)})$, $\ldots$, $Z_n = X_{(n)} - X_{(n-1)}$ are i.i.d. random variables from $\mathrm{Exp}(\lambda)$.
   b) Find $\mathrm{Cov}[X_{(1)} + (n-1)X_{(2)}, \bar{X}_n]$.
   c) Show that

$$\sqrt{n-1}\left(\frac{n\left(\bar{X}_n - X_{(1)}\right)}{n-1} - \frac{1}{\lambda}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{\lambda^2}\right). \tag{43}$$

Advanced Practice Problem II

2. Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables from the Student's t-distribution with 2 degrees of freedom. For any real number x, its probability density is

$$f(x) = \frac{1}{2\sqrt{2}} \left(1 + \frac{x^2}{2}\right)^{-3/2}, \tag{44}$$

which satisfies $\mathrm{E}[X_1] = 0$ and $\mathrm{E}[X_1^2] = \infty$.

a) Show that, for any $t > 0$,

$$P(|X_1| > t) \leq \frac{1}{t^2}. \tag{45}$$

Advanced Practice Problem III

b) Let us further define

$$Y_i = \begin{cases} X_i & (|X_i| \le n^{1/4}) \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad Z_i = \begin{cases} X_i & (|X_i| > n^{1/4}) \\ 0 & \text{otherwise} \end{cases} \tag{46}$$

where $1 \le i \le n$. Then, by definition, $X_i = Y_i + Z_i$. Now, show that for any $\epsilon > 0$,

$$P\left(|\bar{X}_n| \ge \epsilon\right) \le P\left(|\bar{Y}_n| \ge \epsilon/2\right) + P\left(|\bar{Z}_n| \ge \epsilon/2\right). \tag{47}$$

c) Show that, for any $\epsilon > 0$,

$$P\left(|\bar{Y}_n| \ge \epsilon/2\right) \le \frac{4}{\epsilon^2 n^{1/2}}. \tag{48}$$

d) Show that, for any non-negative random variable $W \ge 0$, $\mathrm{E}[W] = \int_0^\infty P(W > t)dt$.

Advanced Practice Problem IV

e) Show that, for any $\epsilon > 0$,

$$P\left(|\bar{Z}_n| \geq \epsilon/2\right) \leq \frac{4}{\epsilon n^{1/4}}. \tag{49}$$

References I

Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.

Ross, S. M. (2020). *A first course in probability*. Pearson Harlow, UK.